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45. $K_C = 4 \times 10^{-2}$ for the reaction,

$$H_2S(g) \rightleftharpoons H_2(g) + \tfrac{1}{2} S_2(g)$$

$$K_C = \frac{[H_2][S_2]^{1/2}}{[H_2S]}$$

$$\Rightarrow 4\times10^{-2} = \frac{[H_2][S_2]^{1/2}}{[H_2S]}$$

For the reaction,

$$2H_2S(g) \rightleftharpoons 2H_2(g) + S_2(g)$$

$$K_C = \frac{[H_2]^2[S_2]}{[H_2S]^2} = \left(4\times10^{-2}\right)^2 = 16\times10^{-4}$$

For the reaction,

$$3H_2S(g) \rightleftharpoons 3H_2(g) + \tfrac{3}{2} S_2(g)$$

$$K_C = \frac{[H_2]^3[S_2]^{3/2}}{[H_2S]^3} = \left(4\times10^{-2}\right)^3 = 64\times10^{-6}$$

46. Given $m_{N_2} = 28$ g $\quad m_{H_2} = 6$g

V = 1 L

$$\left(n_{N_2}\right)_{initial} = \frac{28}{28} = 1 \text{ mol}$$

$$\left(n_{H_2}\right)_{initial} = \frac{6}{2} = 3 \text{ mol}$$

$$N_2(g) + 3H_2(g) \rightleftharpoons 2\,NH_3(g)$$

	$N_2(g)$	$H_2(g)$	$NH_3(g)$
Initial concentration	1	3	–
Reacted	0.5	1.5	–
Equilibrium concentration	0.5	1.5	1

$$[NH_3] = \left(\frac{17}{17}\right) = 1 \text{ mol} = 1 \text{ mol}$$

Weight of N_2 = (no. of moles of N_2) × molar mass of N_2

= 0.5 × 28 = 14 g

Weight of H_2 = (no. of moles of H_2) × molar mass of H_2

= 1.5 × 2 = 3 g

47. $2XY_2(g) \rightleftharpoons 2XY(g) + Y_2(g)$

	XY_2	XY	Y_2
Intial no. of moles	1	-	-
No. of moles dissociated	x	-	-
No. of moles at equilibrium	$(1 - x) \cong 1$	x	$x/2$

Total No. of moles = $1 - x + x + x/2$

$= 1 + x/2 \cong 1$

[$\because$ Given that $x << 1$; $1 - x \cong 1$ and $1 + x/2 \cong 1$]

$$K_P = \frac{(P_{XY})^2 (P_{Y_2})}{(P_{XY_2})^2} = \frac{\left(\frac{x}{1}\times P\right)^2 \left(\frac{x/2}{1}\times P\right)}{\left(\frac{1}{1}\times P\right)^2}$$

$$K_P = \frac{x^2 P^2 \times P}{2P^2}$$

$$\Rightarrow \boxed{2K_P = x^3 P}$$

48. $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

	A_2	B_2	AB
Initial Concentration	1	1	-
No. of moles reacted	x	x	-
No. of moles at equilibrium	1 – x	1 – x	2x

Total no. of moles = 1 – x + 1 – x + 2x

= 2

$$K_P = \frac{(P_{AB})^2}{(P_{A_2})(P_{B_2})} = \frac{\left(\frac{2x}{2}\times P\right)^2}{\left(\frac{(1-x)}{2}\times P\right)\left(\frac{1-x}{2}\times P\right)}$$

$$K_P = \frac{4x^2}{(1-x)^2}$$

Given that $K_P = 1; \frac{4x^2}{(1-x)^2} = 1$